Scatter Plots

Name _____ Date _____ No calculator Version 1

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KEY WORDS scatter plot *(kauwhata marara)* · explanatory variable · response variable

A Explanatory and response variables ●○○

Remember: The explanatory variable is the one that may explain a change in the other one. It goes on the horizontal axis. The response variable goes on the vertical axis.

1. For each pair, say which is the explanatory variable and which is the response variable.

a) Hours of study and test score

b) The temperature outside and the number of ice creams sold

c) The age of a car and the value of the car

d) Hours of training each week and time to run 100 m

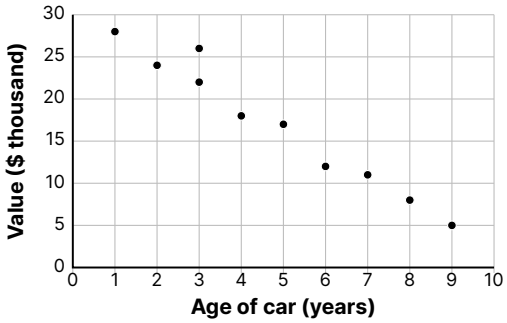
e) The height of a tree and the length of its shadow

f) The amount of rain in a month and the number of umbrellas sold

B Reading a scatter plot ●○○

Remember: Each point shows two measurements for one thing. Read down to one axis and across to the other.

2. The scatter plot shows the ages of some used cars and their values.



a) How many cars are shown? _____

b) What is the value of the 9-year-old car? _____

c) How old is the car with the highest value? _____

d) How many cars are worth less than $15 thousand? ____

e) How many cars are more than 5 years old? _____

f) How many cars are worth more than $20 thousand? ____

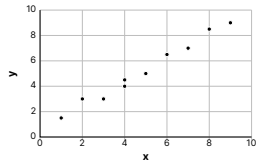
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C Describing the relationship ●●○

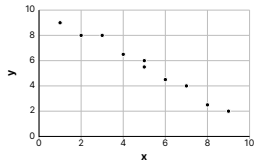
Remember: Positive: as one goes up, so does the other. Negative: as one goes up, the other goes down. The closer the points are to a straight line, the stronger the relationship.

3. Describe the relationship in each scatter plot. Use the words positive or negative, and strong or weak, or write "no relationship".

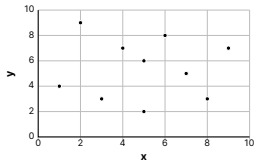
a)



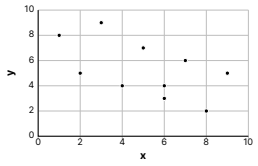
b)



c)



d)

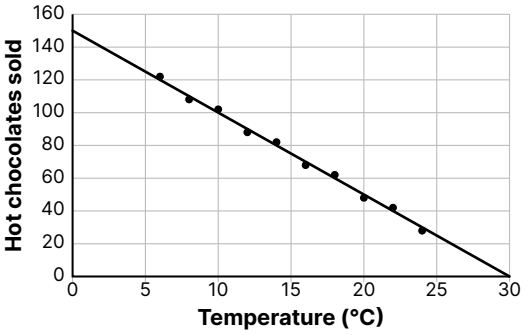


4. Describe the relationship between the age of a car and its value in the scatter plot of used cars above.

D Predicting from a line of best fit ●●●

Remember: A line of best fit follows the trend of the points. To make a prediction, go from one axis to the line and then across to the other axis.

5. A cafe records the temperature and the number of hot chocolates it sells each day. The line of best fit is drawn on the graph.



a) Use the line to predict the number of hot chocolates sold when the temperature is 10°C.

b) Use the line to predict the number of hot chocolates sold when the temperature is 20°C.

c) At what temperature does the line predict that 75 hot chocolates are sold?

d) The line predicts −25 hot chocolates at 35°C. Why does this prediction make no sense?

★ Extension

6. A line of best fit for hours of study h and test score s has the equation $s = 6h + 20$.

a) Predict the test score of a student who studies for 4 hours.

b) Predict the test score of a student who studies for 9 hours.

c) What does the number 20 in the equation tell you?

d) Use the equation to predict the score for 20 hours of study. Why is this prediction not sensible?

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Scatter Plots

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS scatter plot *(kauwhata marara)* · explanatory variable · response variable

A Explanatory and response variables ●○○

Remember: The explanatory variable is the one that may explain a change in the other one. It goes on the horizontal axis. The response variable goes on the vertical axis.

1. For each pair, say which is the explanatory variable and which is the response variable.

a) Hours of practice and music exam mark

b) The temperature outside and the number of hot drinks sold

c) The age of a phone and the value of the phone

d) Hours of training each week and time to swim 50 m

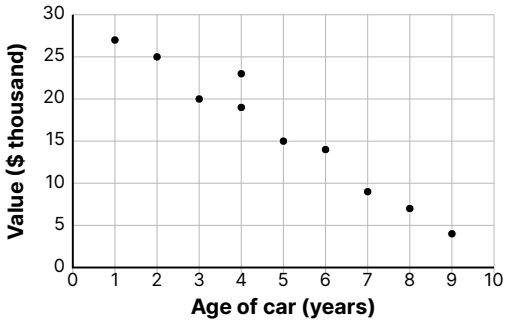
e) The height of a pole and the length of its shadow

f) The hours of sunshine in a day and the number of people at the beach

B Reading a scatter plot ●○○

Remember: Each point shows two measurements for one thing. Read down to one axis and across to the other.

2. The scatter plot shows the ages of some used cars and their values.



a) How many cars are shown? _____

b) What is the value of the 9-year-old car? _____

c) How old is the car with the highest value? _____

d) How many cars are worth less than $16 thousand? ____

e) How many cars are more than 6 years old? _____

f) How many cars are worth more than $20 thousand? ____

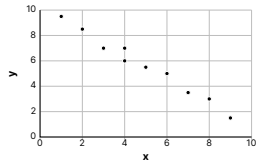
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C Describing the relationship ●●○

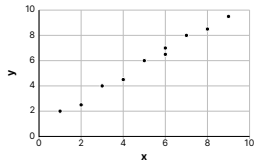
> **Remember:** Positive: as one goes up, so does the other. Negative: as one goes up, the other goes down. The closer the points are to a straight line, the stronger the relationship.

3. Describe the relationship in each scatter plot. Use the words positive or negative, and strong or weak, or write "no relationship".

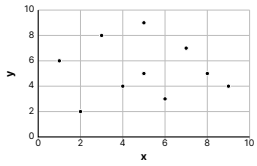
a)



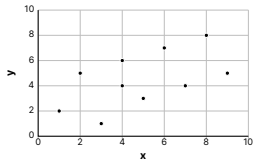
b)



c)



d)

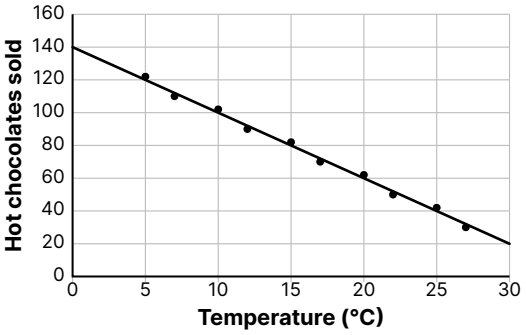


4. Describe the relationship between the age of a car and its value in the scatter plot of used cars above.

D Predicting from a line of best fit ●●●

> **Remember:** A line of best fit follows the trend of the points. To make a prediction, go from one axis to the line and then across to the other axis.

5. A cafe records the temperature and the number of hot chocolates it sells each day. The line of best fit is drawn on the graph.



a) Use the line to predict the number of hot chocolates sold when the temperature is 10°C.

b) Use the line to predict the number of hot chocolates sold when the temperature is 20°C.

c) At what temperature does the line predict that 80 hot chocolates are sold?

d) The line predicts 0 hot chocolates at 35°C. Why does this prediction make no sense?

★ Extension

6. A line of best fit for hours of study h and test score s has the equation $s = 5h + 30$.

a) Predict the test score of a student who studies for 3 hours.

b) Predict the test score of a student who studies for 8 hours.

c) What does the number 30 in the equation tell you?

d) Use the equation to predict the score for 20 hours of study. Why is this prediction not sensible?

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